

# Mid-Term Progress Report

Strategic Coherence in Large Language Models

*Cross-Environment Strategic Profile Analysis via Game Theory (P264)*

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# 1. Executive Summary

This report documents the mid-term progress of the ESADE MIBA 2026 Capstone project investigating whether large language models (LLMs) exhibit stable, transferable strategic dispositions across structurally distinct interaction environments, or whether their behaviour is context-fragmented and prompt-sensitive. The project sits at the intersection of behavioural game theory, AI evaluation methodology, and computational social science.

By the midpoint, our team has accomplished three substantive milestones. First, a formal conceptual framework designed and approved by our project supervisor, defining a strategic profile vector  $S_m = (\rho, \theta, \eta, \gamma, \beta)$  that captures five behavioural parameters across three structurally distinct game-theoretic environments. Second, a production-quality multi-model experimental pipeline was built in Python using LangChain, supporting Claude, GPT-4o, and Gemini model families through a single unified interface. Third, pilot experiments were run on the Iterated Prisoner's Dilemma, yielding 40 rounds of clean behavioural data - including actions, stated beliefs, payoffs, and response times, that produce theoretically consistent results.

The remaining work centres on scaling the Prisoner's Dilemma experiments across model families and temperature settings, implementing the Commons Dilemma and Cheap-Talk Signaling environments, and conducting the full cross-environment coherence analysis that forms the paper's core contribution.

## 2. Project Overview

### 2.1 Research Question

The central question driving this project is whether large language models possess coherent strategic architectures; stable, transferable behavioural dispositions that manifest consistently across different types of strategic interactions, or whether their behaviour is simply a function of the immediate prompt context, producing different strategies in different games without any underlying coherence. This question matters both scientifically, as a contribution to our understanding of what LLMs actually “know” about strategy, and practically, as AI systems are increasingly deployed in negotiation, resource allocation, and multi-agent settings where strategic consistency is relevant to trust and governance.

The empirical strategy is to estimate formal behavioural and belief parameters within each of three game environments and then test whether these parameters form a coherent strategic profile at the model level. Three environments were selected for their structural contrast: the Repeated Prisoner's Dilemma (dyadic reciprocity under intertemporal incentives), the Commons Dilemma (collective action under resource congestion), and a Cheap-Talk Signaling game (strategic communication without binding commitments). Together these three environments activate meaningfully different strategic logics, making them a strong basis for testing coherence.

## 2.2 Conceptual Framework

Each model  $m$  is characterised by a strategic profile vector  $S_m = (\rho_m, \theta_m, \eta_m, \gamma_m, \beta_m)$ . Conditional reciprocity ( $\rho$ ) captures the degree to which a model's cooperation is contingent on its opponent's prior action in the Prisoner's Dilemma, defined as the difference in cooperation probability after the opponent cooperated versus after the opponent defected. Exploitation threshold ( $\theta$ ) is the pool size in the Commons game at which the model's extraction first exceeds the sustainable regeneration rate, measuring the transition from restraint to over-extraction. Signal honesty ( $\eta$ ) is the fraction of rounds in which a model's sent message matched its intended action in the Cheap-Talk game. Receiver gullibility ( $\gamma$ ) measures how strongly a model's cooperation decision is influenced by the sender's message, regardless of its accuracy. Belief calibration ( $\beta$ ) is the mean absolute error between stated cooperation probabilities and actual opponent actions, elicited before each round across all three environments.

Three hypotheses structure the analysis: H1 (Scaling) predicts that larger models exhibit more stable strategic profiles across environments. H2 (Instruction tuning) predicts that instruction-tuned models show higher norm-conforming cooperation but also potentially higher gullibility. H3 (Temperature) predicts that higher sampling temperatures increase behavioural variance and reduce cross-environment coherence. A perturbation test compares each model's profile under AI-AI matchups versus human opponents, with large deviations indicating opponent-dependent strategy rather than robust architecture.

## 3. Goals

The primary research objective is to produce the first unified cross-environment strategic profile analysis of LLMs: estimating the five formal parameters for each model, constructing the profile vector, and computing cross-environment coherence metrics including variance ( $C_m = \text{Var}(S_m^{\text{env}})$ ), parameter correlations, and a latent factor structure. The analysis will

test all three hypotheses and produce human-calibrated interpretations using existing literature benchmarks as reference distributions.

Beyond the research contribution, the project serves clear learning objectives. Team members gain practical experience in experimental design with live LLM APIs, applied behavioural game theory, and AI evaluation methodology analogous to Red Teaming at OpenAI, Google DeepMind, and Anthropic. The end deliverable is a complete research paper suitable for an academic venue, alongside the open-source experimental codebase.

## 4. Work Done to Date

### 4.1 Conceptual Framework

In March 2026, the team designed the full formal conceptual architecture from scratch: the strategic profile vector definition, the operational measurement procedures for each parameter, the three coherence metrics, and the three testable hypotheses. This document was submitted to the project supervisor, Carlos Carrasco-Farré, and approved as the primary direction to pursue. The framework has each parameter defined in terms of specific measurable quantities derivable from round-by-round action and belief data, not just conceptual labels.

### 4.2 Experimental Infrastructure

We built a 659-line production-quality experimental pipeline in Python, hosted at ([github.com/gabrielamendez22/Capstone\\_2026\\_Game\\_Theory](https://github.com/gabrielamendez22/Capstone_2026_Game_Theory)). The central design decision was to use LangChain as a unified interface layer, so that Claude (Anthropic), GPT-4o (OpenAI), and Gemini (Google) can all be called through the same `.invoke()` method. Adding a new model family requires changing a single line. This is a meaningful engineering choice: prior LLM game theory papers typically hard-code one provider, making cross-family comparisons cumbersome. The registry currently supports Claude Opus and Sonnet, GPT-4o and GPT-4o-mini, and Gemini 1.5 Pro and Flash, with six pre-configured matchups.

The game loop implements several features important for the strategic profile analysis. Belief elicitation is built into every round: before choosing an action, each model is prompted to state its probability that the opponent will cooperate, producing the data needed for  $\beta$  (belief calibration) measurement. The payoff matrix uses standard PD values ( $T=5$ ,  $R=3$ ,  $P=1$ ,  $S=0$ ), satisfying the canonical constraint  $T > R > P > S$ . History is passed to each model as a structured round-by-round log, with a configurable sliding window to test whether models behave differently under limited versus full memory. Data is written to SQLite after every round for crash safety, and exported to CSV for statistical analysis. Prompt versioning is

tracked (v1.0 in the pilot, v3.0 in the current production build) to support reproducibility. A separate two-bot conversation infrastructure using the Cohere API was also built for earlier dialogue experiments.

## 4.3 Pilot Experiments

On March 27, 2026, the team ran two pilot games of the Iterated Prisoner's Dilemma: Gemini Flash versus Gemini Flash (self-play), and Gemini Pro versus Gemini Flash (cross-model). Each game ran for 20 rounds at temperature 0.7 using prompt version v1.0. Results were logged to `pd_results_20260327_100443.json` and the corresponding CSV file.

# 5. Pilot Results - Prisoner's Dilemma

## 5.1 Overview

The two pilot games produced 40 rounds of behavioural data. In the Gemini Flash self-play (Game 1), both instances cooperated in 19 of 20 rounds, yielding a 95% cooperation rate on each side and a symmetric final score of 58 points each. In the Gemini Pro versus Gemini Flash matchup (Game 2), cooperation rates were 90% and 85% respectively, with a final score of 53 for Pro and 58 for Flash. The score asymmetry arose from a single strategic episode in rounds 18 to 20.

## 5.2 Belief Calibration

Belief calibration ( $\beta$ ), computed as the mean absolute error between each model's stated cooperation probability and the opponent's actual action, was remarkably low across both games. Gemini Flash showed  $\beta$  values of 0.090 (as model A in Game 1) and 0.071 (as model B in Game 1), with similar values of 0.090 in Game 2. Gemini Pro showed  $\beta = 0.095$  in Game 2. These values are close to zero, indicating near-Bayesian belief updating: the models' stated probabilities closely tracked actual opponent behaviour throughout each game. For reference, a  $\beta$  of 0.5 would indicate predictions no better than random, and  $\beta$  of 0 would be perfect calibration.

The belief trajectories are particularly revealing. Both models started with neutral or moderately optimistic priors (0.50 to 0.70 in round 1) and updated rapidly upward through sustained cooperation, reaching confidence levels of 0.98 to 0.99 by rounds 10 to 15. When defection occurred, beliefs crashed immediately to 0.05, indicating near-certain expectation of continued defection. This pattern is consistent with Bayesian updating under a simple model of opponent behaviour.

## 5.3 Cooperation Dynamics and End-Game Defection

In Game 1 (Flash self-play), cooperation was sustained perfectly for 19 consecutive rounds before both instances defected simultaneously in round 20. This is consistent with the game-theoretic prediction of end-game defection in finitely repeated games: when both players know the game is about to end, the incentive to defect for a one-round gain outweighs the cost of breaking cooperation. The fact that both instances defected in the same final round - despite being called independently - suggests a shared internal model of the game's structure.

Game 2 (Pro vs Flash) showed a more interesting pattern. Cooperation held for 17 rounds, but Gemini Flash began showing declining beliefs from round 16 onward (dropping from 0.95 to 0.90 in round 16, then to 0.80 in round 17), before defecting first in round 18. Gemini Pro, still holding a belief of 0.99 in round 18, was caught off guard and received the sucker's payoff ( $S=0$ ) while Flash received the temptation payoff ( $T=5$ ). Pro then retaliated with defection in rounds 19 and 20. This asymmetric defection cascade - where one model anticipates the end of cooperation before the other - is theoretically interesting: it suggests the two models have different internal "end-game horizons" and raises questions about whether this difference is a stable property of the model architecture or a product of temperature and prompt sensitivity.

## 5.4 Early Observations on Strategic Parameters

Both models display high baseline cooperation rates (90 to 95%) and strong belief calibration ( $\beta \approx 0.09$ ), but the pilot data does not yet contain enough defection events to robustly estimate conditional reciprocity ( $\rho$ ). With only end-game and cascade defections observed, the difference  $P(C \mid \text{opponent cooperated}) - P(C \mid \text{opponent defected})$  cannot be estimated with statistical confidence. This is a known limitation of high-cooperation pilot conditions and will be addressed in the full experimental grid by introducing a wider range of model pairings, higher temperatures, and adversarial bot conditions that force more defection variation. The Commons Dilemma and Cheap-Talk environments are structurally designed to produce more behavioural variation and will contribute the remaining three parameters ( $\theta, \eta, \gamma$ ).

## 6. Work to Be Done

The immediate next step is scaling the Prisoner's Dilemma experiments across the full model grid. This means running all six planned matchups (three cross-family pairings and three same-family size comparisons) at three temperature levels (approximately 0.1, 0.7, and 1.0) to test H3, and comparing base versus instruction-tuned variants where available

to test H2. This scale-up will also generate sufficient defection variation to robustly estimate  $p$  across models.

In parallel, the team will implement the Commons Dilemma. This requires building a multi-round resource extraction game in which a shared pool regenerates at a fixed rate and each model chooses how much to extract per round. The exploitation threshold  $\theta$  will be estimated as the pool size at which each model's extraction first exceeds the sustainable share. Like the PD, this will run across the full model-temperature-tuning grid.

The Cheap-Talk Signaling game follows. This involves designing a game with incentive misalignment between sender and receiver, logging each model's intended action before its message is generated (the key step for measuring  $\eta$ ), and measuring how strongly the receiver's action shifts in response to messages of different types (the basis for  $\gamma$ ). The "low  $\eta$  sender, high  $\gamma$  receiver" pairing is theoretically the most interesting outcome to look for, as it represents an exploitable strategic asymmetry.

Once all three environments are complete, the analysis proceeds in four steps: computing the profile vector  $S_m$  per model, computing cross-environment variance  $C_m$  as the primary coherence metric, running correlation and latent factor analyses to test structural coherence, and computing the perturbation metric  $\Delta_m = ||S_m^{AI} - S_m^H||$  using published human benchmarks. The findings will then be synthesised into an academic paper.

## 7. Added Value

For the team, this project provides hands-on experience with the full cycle of empirical AI research: from formal hypothesis design through experimental implementation, data collection, and statistical analysis. The skills developed - LLM API engineering, prompt design, behavioural game theory, and cross-model evaluation - are directly analogous to the Red Teaming work conducted at OpenAI, Google DeepMind, and Anthropic, and represent a rare practical exposure for an MBA programme.

For the academic community, the project fills a clear and identified gap. Existing work on LLMs in game theory studies one environment at a time, compares AI to humans, or focuses on a single model family. What is missing - and what this project delivers - is a unified analysis that constructs a formal model-level strategic profile, tests cross-environment coherence explicitly, and runs a systematic AI-AI grid across scale, tuning, and temperature. The supervisor has confirmed this framing as the cleanest and most defensible interpretation of the research question, with strong publication potential.

For society, the findings speak directly to how AI systems should be trusted and governed in strategic contexts. As LLMs are deployed in negotiation support, supply chain coordination,



multi-agent autonomous systems, and other settings where consistent strategic behaviour matters, knowing whether a model's cooperation in one context predicts its cooperation in another is practically significant. The project contributes to the empirical evidence base for responsible AI deployment.

Relevant UN Sustainable Development Goals: SDG 9 (Industry, Innovation and Infrastructure) through the development of novel AI research methodology; SDG 16 (Peace, Justice and Strong Institutions) through contributions to responsible AI governance and transparency; and SDG 17 (Partnerships for the Goals) through cross-disciplinary collaboration across computer science, behavioural economics, and ethics.

## 8. Risks and Mitigations

Technical risk: API rate limits, model deprecation, or inconsistent outputs across runs could affect reproducibility and data quality. The pipeline mitigates this through round-by-round SQLite persistence (data survives crashes), timestamped CSV exports, and prompt versioning so that any re-run can be traced to its exact configuration.

Methodological risk: Prompt sensitivity may confound strategic profile estimation: a model's apparent cooperation rate could reflect the framing of the system prompt rather than a stable underlying disposition. The project addresses this directly by treating prompt variance as a measured property rather than noise - the same game will be run under multiple prompt versions, and variance across frames is itself an input to the coherence analysis.

Scope risk: Running three environments across six or more model variants, three temperatures, and multiple tuning conditions is computationally and financially non-trivial. The mitigation strategy is to prioritise the Prisoner's Dilemma full grid first (since it is already implemented), and to use a reduced but representative model set for the Commons and Cheap-Talk environments if time or budget becomes constrained.

Data quality risk: The pilot showed very low defection rates (only end-game), which limits  $p$  estimation. The full grid will address this through wider model pairings (some inherently more defection-prone), higher temperatures, and the inclusion of adversarial bot opponents (e.g. always-defect) to force sufficient behavioural variation for robust parameter estimation.

## 9. What We Are Most Proud Of

As a team, we are most proud of the fact that the conceptual framework was designed from first principles and approved by the supervisor on the first submission. Starting from a broad

research question about AI strategic behaviour, we translated it into a formal, operationalised measurement framework with precise parameter definitions - something that required both theoretical clarity and methodological discipline.

We are also proud of the experimental infrastructure. Building a 659-line, multi-provider LLM pipeline with crash-safe data logging, belief elicitation, and configurable history windows is a genuine technical achievement for an MBA team, and the results already show behaviour that is theoretically interpretable: the end-game defection in Game 1 and the asymmetric defection cascade in Game 2 are exactly the kinds of patterns game theory would predict, which gives us confidence in the pipeline's validity.

## 10. GitHub Repository

The full experimental codebase is publicly available at ([github.com/gabrielamendez22/Capstone\\_2026\\_Game\\_Theory](https://github.com/gabrielamendez22/Capstone_2026_Game_Theory)).

The repository is written entirely in Python and contains four key files. The main experiment script (`prisoners_dilemma_langchain.py`, 659 lines) implements the unified LangChain interface, the full game loop, belief elicitation, SQLite logging, and CSV export. The pilot results file (`pd_results_20260327_100443.csv`) contains 40 rounds across the two Gemini matchups, with columns for actions, beliefs, payoffs, cumulative scores, raw model outputs, and response times. An environment specification (`environment.yml`) and requirements file (`requirements.txt`) ensure full reproducibility of the Python environment.